

THE SPRITE MODULE

Pygame has a built-in module to show images in a window, called the *sprite module*. Such images are called *sprites*. The sprite module provides a `Sprite` class for handling individual sprites and a `Group` class for handling multiple `Sprite` objects. Together, these classes provide excellent functionality, and if you intend to do heavy-duty pygame programming, it is probably worth your time to look into them. However, in order to explain the underlying OOP concepts, I have chosen not to use those classes. Instead, I will proceed with general GUI elements so that they can be used in any environment and language. If you want to learn more about the sprite module, see the tutorial at <https://www.pygame.org/docs/tut/SpriteIntro.html>.

Adding Buttons, Checkboxes, and Radio Buttons

When you instantiate a button, checkbox, or radio button widget in `pygwidgets`, you have two options: instantiate a text version that draws its own art and adds a text label based on a string you pass in, or instantiate a custom version where you supply the art. [Table 7-1](#) shows the different button classes that are available.

Table 7-1: *Text and Custom Button Classes in `pygwidgets`*

	Text version (builds art on the fly)	Custom version (uses your artwork)
Button	<code>TextButton</code>	<code>CustomButton</code>
Checkbox	<code>TextCheckBox</code>	<code>CustomCheckBox</code>
Radio button	<code>TextRadioButton</code>	<code>CustomRadioButton</code>

The differences between the text and custom versions of these classes are only relevant during instantiation. Once you create an object from a text or custom button class, all the remaining methods of the pair of classes are identical. To make this clear, let's take a look at the `TextButton` and `CustomButton` classes.

TextButtons

Here is the actual definition of the `__init__()` method of the `TextButton` class in `pygwidgets`: